

When Reading an Equivariance Monitor’s Residual Through the Model Helps, and When It Hurts: A Locality Law from Three Sealed Preregistrations

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Abstract

An equivariance monitor grades a paraphrase-invariance residual $e = h_\ell(gx) - \hat{\rho}_\ell h_\ell(x)$ in raw L2 norm. We ask whether grading it instead in the downstream model’s own read, measured as the model’s output response when e is patched into layer ℓ , catches more real behavioral changes at matched flag rates. The answer is conditional, and we pin the condition through three preregistered families, two of which failed their own sealed bars and are reported as executed. The consumer read cuts monitor false-clears by 14 to 21 percentage points under backtranslation, but under paraphrase it does the opposite, adding 35 to 57 points, at two model scales. Two sealed attempts to gate the effect on the equivariance map’s own fit both failed. The variable that separates the regimes, identified from the failures and then preregistered as a law, is transform locality: the held-out R^2 of the identity map $h_\ell(gx) \approx h_\ell(x)$. Where the transform is local in representation space the patched residual points along the true perturbation and the consumer read is informative even when the equivariance map is a poor fit; where it is not, the read is dominated by map error and inverts. The practical rule is a mechanical, calibration-only gate: below a locality floor a monitor must fall back to the raw metric or abstain.

1 Introduction

Internal-equivariance monitors gate on $h_\ell(gx) \approx \hat{\rho}_\ell(g) h_\ell(x)$: a paraphrase g should move a layer’s representation the way a fitted linear map $\hat{\rho}_\ell$ predicts, and a residual that violates this flags a possible spec-gaming or drift. The residual is scored in raw L2 norm. Observation Theory raises an objection that applies to any such monitor: the layer’s activations have a consumer, the downstream computation, and the norm that matters is the consumer’s read of the residual, not its raw magnitude. A large residual in a direction the downstream network ignores is a false alarm; a small residual exactly where the network reads is a false clear, which is the case the monitor exists to catch.

This paper measures the consumer-read alternative and finds it conditional, in a way that is itself the contribution. We grade the residual by its measured patch-response: inject e at layer ℓ and record the model’s output-KL response, the consumer’s actual sensitivity to that residual. Across three preregistered families we find a clean advantage under one transform family, a clean inversion under another, and, after two sealed attempts to predict which is which from the wrong variable, a locality law that does predict it. The methodological spine is preregistration: pass/fail thresholds and seeds frozen in version control before evaluation, verdicts committed as executed.

Two of the three families failed, and both failures are load-bearing evidence for the law the third family states.

2 Method

Setup. An open causal LM (Qwen2.5, 0.5B and 1.5B) reads Social-Chemistry-101 action texts and their transforms. For each layer ℓ the whitepaper’s ridge estimator fits $\hat{\rho}_\ell$ on a calibration split; the residual e is injected at ℓ by a last-position activation patch and the metric is the model’s output-KL response. Ground truth is the model’s actual next-token KL between x and gx . The monitor flags the top quartile by each metric; the reported quantity is the false-clear rate, the fraction of behavior-changed pairs not flagged, at matched flag rates, and the advantage $\Delta = f_{c_{\text{raw}}} - f_{c_{\text{true}}}$.

Discipline. Each family is a sealed preregistration naming its cells, bars, seeds, and manipulation checks before any evaluation data is opened, with a one-day cooling-off between construction and seal. Graded verdicts are committed exactly as the frozen rules grade them. All records, seals, and post-mortems are in the campaign repository.

3 The effect is real, and it inverts by transform family

The construction cells established the shape at shakedown grade: under backtranslation, the consumer read of the residual predicts behavioral change better than raw L2 and cuts false-clears by 15 to 21 points at matched flag rates, replicated on a second model (TinyLlama-1.1B) and a second backtranslation language. The final-layer equivariance map fails to generalize in every run, held-out R^2 collapsing from above 0.8 in calibration to at or below zero, so a monitor calibrated at the final layer grades against a broken map. Both findings held under seal (family V1): the sealed backtranslation cell cut false-clears, and the final-layer collapse recurred.

Family V1 nonetheless graded FAIL on its quantitative bar, and its post-mortem found the effect’s magnitude unforecastable from the crude single-split headroom it had qualified on. Family V2 repaired the instrument and introduced paraphrase as a second transform family. The result was decisive in an unexpected direction: under paraphrase, at both model scales, the consumer read did not merely lose its advantage, it inverted, adding 35 to 57 points of false-clears over raw L2. The mechanism is visible in the records: paraphrase drives the equivariance map’s held-out R^2 to zero or below, so the patched residual is dominated by map error, and injecting map error probes the model’s sensitivity to a direction that is not the perturbation.

4 Two sealed attempts to gate the effect on map fit both fail

The natural gate is the equivariance map’s own quality: use the consumer read only where $\hat{\rho}$ generalizes. Family V2’s sealed secondary and the V2 successor both tested a version of this, gating on the held-out R^2 of $\hat{\rho}$. It fails, and the failure is precise. In the deep layers of the 1.5B backtranslation cells the map’s held-out R^2 is slightly negative, so a $\hat{\rho}$ -fit gate calls them out-of-regime, yet the consumer advantage persists there at 9 to 17 points. The advantage does not need a good map; it needs the patched residual to point roughly along the true perturbation. Map fit is a proxy for that and a leaky one.

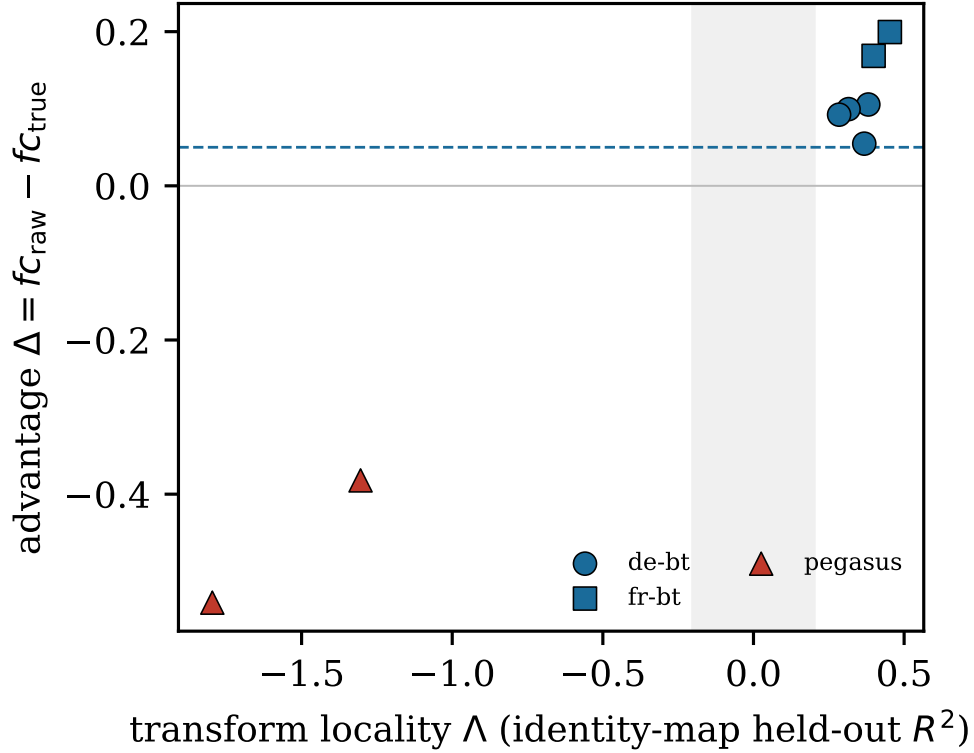


Figure 1: The consumer-read advantage Δ against transform locality Λ , the identity-map held-out R^2 , one point per graded (cell, layer) unit. Backtranslation units (high Λ) sit above the advantage line; paraphrase units (negative Λ) invert. The shaded band is the preregistered transition region. [V3 synonym/shuffle/truncate units append here when graded.]

5 The locality law

The variable that separates the regimes, read directly off the failures, is transform locality: how nearly $h_\ell(gx) \approx h_\ell(x)$, measured as the held-out R^2 of the identity map,

$$\Lambda_\ell = 1 - \frac{\sum \|h_\ell(gx) - h_\ell(x)\|^2}{\sum \|h_\ell(gx) - \overline{h_\ell(gx)}\|^2}.$$

It requires no map fit. On the V2 records it separates the outcomes with no overlap: every backtranslation unit, including the two the $\hat{\rho}$ -gate got wrong, has Λ between $+0.28$ and $+0.45$; both paraphrase units have Λ at -1.3 and -1.8 . When a transform is local, $h_\ell(gx) = h_\ell(x) + \delta$ with small δ , so even a noisy $\hat{\rho}$ leaves the residual approximately equal to the true perturbation and the consumer read stays informative; when the transform relocates the representation wholesale, the residual is map error and the read inverts.

Family V3 preregisters this as a law rather than a threshold. It grades cells spanning a constructed locality ladder, from synonym substitution ($\Lambda \approx 0.9$) through interior word-shuffling ($\Lambda \approx 0.4$, textually very different yet representationally local, the case that distinguishes locality from map fit) to half-truncation ($\Lambda \approx -0.7$). The sealed predictions are a threshold law, $\Delta \geq 0.05$ when $\Lambda \geq 0.20$ and $\Delta \leq 0.01$ when $\Lambda \leq -0.20$, and a monotonicity, $\text{Spearman}(\Lambda, \Delta) \geq 0.7$ pooled

over all units. [Section to complete on the V3 graded record; Figure 1 shows the V1 and V2 units the law was read from.]

6 Consequence for monitor design

The rule the three families support is operational and cheap. A consumer-metric equivariance gate must be conditioned on a mechanical, calibration-only locality check: compute Λ on a held-out slice of the calibration split, and below a locality floor fall back to the raw metric or abstain, because the consumer read there is worse than the raw metric it would replace. This is the $\hat{\rho}$ -freshness and held-out-validation requirement of the whitepaper, sharpened: it is not enough that $\hat{\rho}$ was fit; the transform the monitor faces must be local enough for the read to be informative, and that is measurable before deployment without fitting $\hat{\rho}$ at all. The two failed paraphrase families are the measured demonstration of the cost of skipping the check.

7 Scope and limitations

Two model scales of one family, Social-Chemistry text, backtranslation and two paraphrasers plus the V3 string-operation ladder, the consumer read measured in x 's context. The final-layer map collapse and the paraphrase inversion are robust across every run measured; the locality law's monotonicity claim is under seal at submission and reported as graded. We claim nothing about transforms outside the measured locality range, and nothing about monitors whose consumer is an external evaluator rather than the model's own output, which the read operator would have to be re-derived for.

Reproducibility

Every number is read from committed, hashed records in the campaign repository (three sealed preregistrations, per-cell graded JSONs, post-mortems, the locality gate and its checksum). Figure 1 regenerates from them by script.